Unit 1: Worked solutions

> Reference: Jennifer Rupp, Daniele Ielmini, Ilia Valov (eds.), Resistive Switching: Oxide Materials, Mechanisms, Devices and Operations, Springer, 2022. DOI: 10.1007/978-3-030-42424-4.

E01

$G_1 = 2/0.1 = 20\,\mu S$ and $G_2 = 6/0.2 = 30\,\mu S$. The different ratios exclude a fixed-conductance, origin-passing linear model over these points; they do not uniquely identify a conduction mechanism. Record bias and polarity, pulse duration and timing, temperature, programmed state, compliance and instrument conditions.

E02

SET increases conductance; RESET decreases it. Forming is a device-dependent initialization, and read measures the state while requiring a disturbance check. Under the stated bipolar convention RESET normally has negative polarity. Different stacks need not share this sign. Current compliance limits excessive conduction and heating, affects the final state, and does not guarantee immunity from failure.

E03

No: hysteresis does not uniquely identify a mechanism. ECM commonly involves mobile metal cations and redox; VCM involves oxygen ions or vacancies and local valence changes. Electrode substitution, isotope tracing, local chemistry or in-situ microscopy can provide evidence. Control field, electrodes and environment, and account for limited spatial resolution, sample preparation and probe-induced changes; correlation alone need not establish causation.

E04

Column 1 branches carry $2(0.2) = 0.4$ and $1(0.1) = 0.1\,\mu A$, totaling $0.5\,\mu A$. Column 2 branches carry $1(0.2) = 0.2$ and $3(0.1) = 0.3\,\mu A$, also totaling $0.5\,\mu A$. At the stated actual second-row crosspoint voltage, the totals become $0.4 + 0.08 = 0.48\,\mu A$ and $0.2 + 0.24 = 0.44\,\mu A$. Linear branch equations remain valid using actual local voltages, but the input terminal voltage 0.1 V cannot be substituted at every second-row crosspoint. Generally wire resistance makes local voltages unequal, requiring a circuit solution.

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Unit 1: Worked solutions (continued)

E05

The defined margin is $5 - 3 = 2\,\mu A$. With the stated strict decision rule, every $3 \leq T < 5\,\mu A$ separates the intervals. Other equality conventions change endpoint inclusion. Overlap precludes a guarantee of zero error from the ranges alone; error rate needs state distributions, prior probabilities, threshold and measurement noise.

E06

Report programming amplitude and duration, current compliance, read voltage and timing, temperature, failure criterion, sample count and structure. Cycle-to-cycle variability describes repeated operations on one device; device-to-device variability compares devices. A single accelerated-test point does not observe ten years directly. Extrapolation needs a validated failure model, fitted parameters and uncertainty, and must consider possible changes of failure mechanism.

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